Analytic parameter sensitivities in ManifoldsX: scope and validation

manifoldsx.github.io/differentiable-dynamics

ManifoldsX computes how motion and joint loads change with spring and bushing stiffness and damping, analytically, at every step of a simulated trajectory. The derivatives cover consistent initialization, the implicit index-3 HHT step and the rotation update. They were compared with independent perturbed simulations over 30-second runs. These are numerical checks of the derivatives; they say nothing about hardware accuracy.

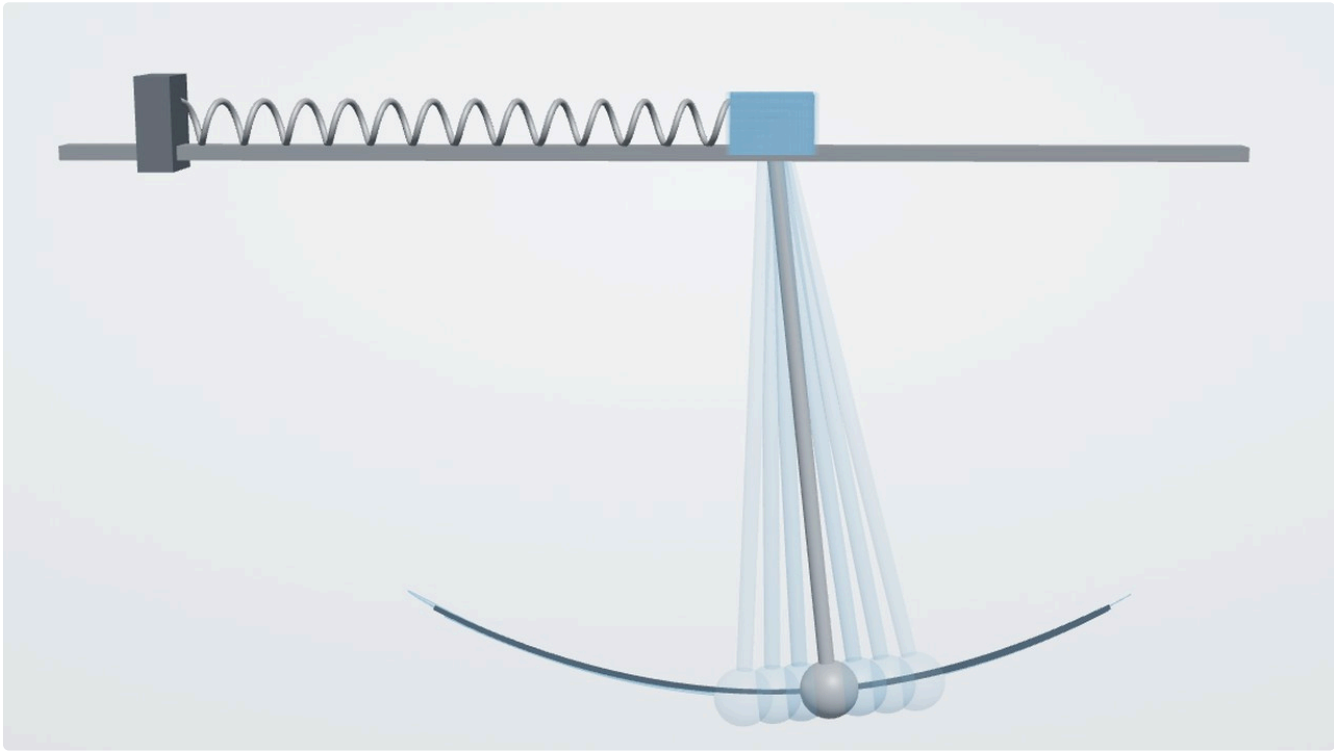


Figure 1. The slider-pendulum-spring test model. The solid pendulum is the simulated motion; the faint copies show how it shifts when the spring stiffness changes, predicted from the analytic gradients of a single run.

Background

“What if this part were stiffer?” is one of the oldest questions in machine design, and the usual way to answer it is to change the number and run the simulation again. Computing the derivative directly is the more efficient route. Forward sensitivity equations follow the derivative of the state alongside the state itself, while adjoint methods run backwards from a quantity of interest. Both are well established for differential-algebraic equations in numerical analysis [1]. They are also what makes a simulator “differentiable”: calibration, design optimization and learning methods can follow a gradient instead of guessing.

For a model of a mechanism the derivative has to pass through everything the solver does in a step. That includes the consistent initial state, the implicit index-3 HHT solve with its Newton iterations [2], [3], and the update of rotations, which do not add like ordinary vectors. Finite differences avoid that work, but they need an extra simulation for every parameter, and their accuracy depends on a perturbation size that has to be guessed. Here they serve only as the reference, extrapolated to remove most of their own error.

1. What they are for

Understanding compliance

How a stiffness or damping coefficient changes displacement, velocity, acceleration and reaction forces at every instant.

Calibration and design

Gradients are the building blocks for parameter fitting and design studies. A fitting or optimization workflow for a given application still has to be built and validated.

2. Validation

Table 1. Validation results

Study	Evidence
Spring stiffness and damping	Analytic trajectory derivatives compared with Richardson-extrapolated finite differences over 30-second runs; every closed-form term also checked against finite differences.
6D bushings	All 12 diagonal coefficients; 24 trajectories of 30 seconds on grounded and moving-parent connections.
Bushing motion derivatives	Worst sampled error relative to the peak reference: 1.11×10^{-7} for position and rotation, 2.37×10^{-7} for velocity.
Bushing accelerations and loads	Same metric: 3.80×10^{-5} for acceleration, 5.47×10^{-5} for multipliers and 5.46×10^{-5} for generalized reactions. These are limited by the finite-difference reference, not by the derivatives.
Sparse sensitivity solve	Damped chains from 10 to 500 links, with one symbolic pattern analysis per run.

Errors are normalized by the peak reference value over the sampled times, not per component at every instant. The analytic runs used no finite-difference evaluations; independent perturbed simulations supplied the reference. In the bushing tests, computing sensitivities left the nominal trajectory bit-for-bit unchanged.

3. Scope

Covered

Forward sensitivities for spring stiffness and damping and the 12 diagonal bushing coefficients, one scalar parameter per sensitivity instance; bushing support is opt-in. The trajectory studies use index-3 HHT with full-rank revolute, spherical and translational joints and a parameter-independent initial pose and velocity.

Not covered yet

Contact, controllers and runtime loads, redundant or rank-changing constraints, projection and other integrators. Adaptive step decisions are not differentiated. The Stewart motion benchmark does not establish differentiability of that whole model.

References

[1] Y. Cao, S. Li, L. Petzold and R. Serban, “Adjoint sensitivity analysis for differential-algebraic equations: the adjoint DAE system and its numerical solution,” *SIAM Journal on Scientific Computing*, vol. 24, no. 3, pp. 1076–1089, 2003.

[2] H. M. Hilber, T. J. R. Hughes and R. L. Taylor, “Improved numerical dissipation for time integration algorithms in structural dynamics,” *Earthquake Engineering and Structural Dynamics*, vol. 5, pp. 283–292, 1977.

[3] D. Negrut, R. Rampalli, G. Ottarsson and A. Sajdak, “On an implementation of the Hilber–Hughes–Taylor method in the context of index 3 differential-algebraic equations of multibody dynamics,” *Journal of Computational and Nonlinear Dynamics*, vol. 2, no. 1, pp. 73–85, 2007.